

An Analysis of Atmospheric Moisture Variables for Added Accuracy in Hail Forecasting



Preston Beard¹, Jonathan Douglas^{1,2}, Ethan Stroberg^{1,2}, Liam Thompson¹, Cameron Homeyer¹, Cameron Nixon^{2,4}, & Sean Waugh³

¹School of Meteorology, University of Oklahoma, Norman, OK, USA, ²Cooperative Institute for Severe and High-Impact Weather Research and Operations, University of Oklahoma, Norman, OK, USA, ³NOAA National Severe Storms Laboratory, National Weather Center, Norman, OK, USA, & ⁴NOAA Storm Prediction Center, National Weather Center, Norman, OK, USA.

Contact: Jonathan.c.douglas-1@ou.edu



i) Background

- Methods to create **accurate** hail forecasts remain far from perfect
- Distinction between *hail forecasting* & *hail detection*
- Modeled variables, including CAPE and shear, show complex relationships for hail production
- Our study seeks to understand the role parameters including ECAPE could play in aiding the **creation of accurate, operational hail forecasts**

ii) Datasets

- Homeyer et al. 2023 – ERA5 reanalysis
- Nixon et al. 2024 (1a-2d) – 20-years of hail observations

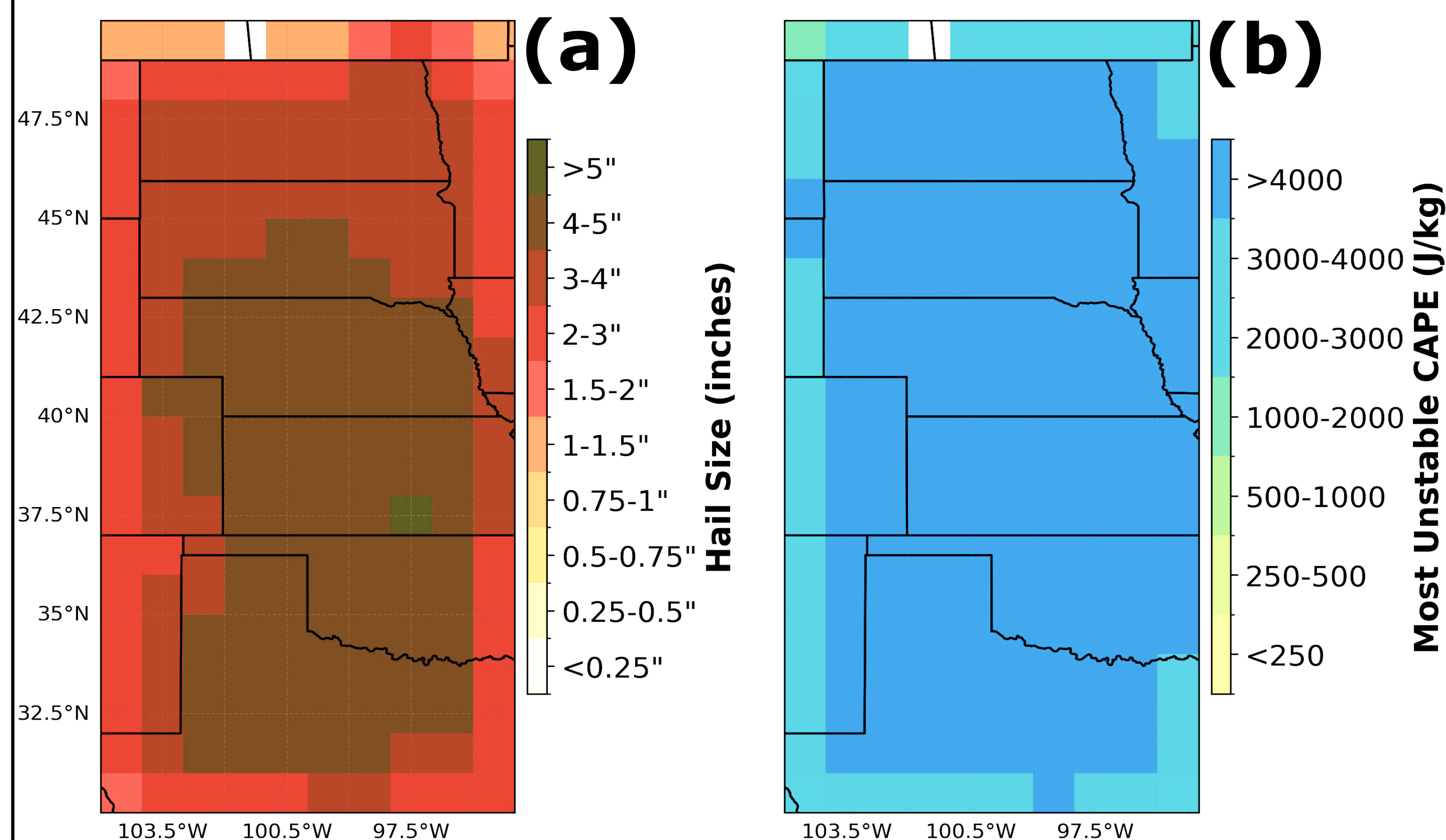


Fig. 1: 2003-2019 Average of Max Hail Size (a) vs Max Most Unstable CAPE (b)

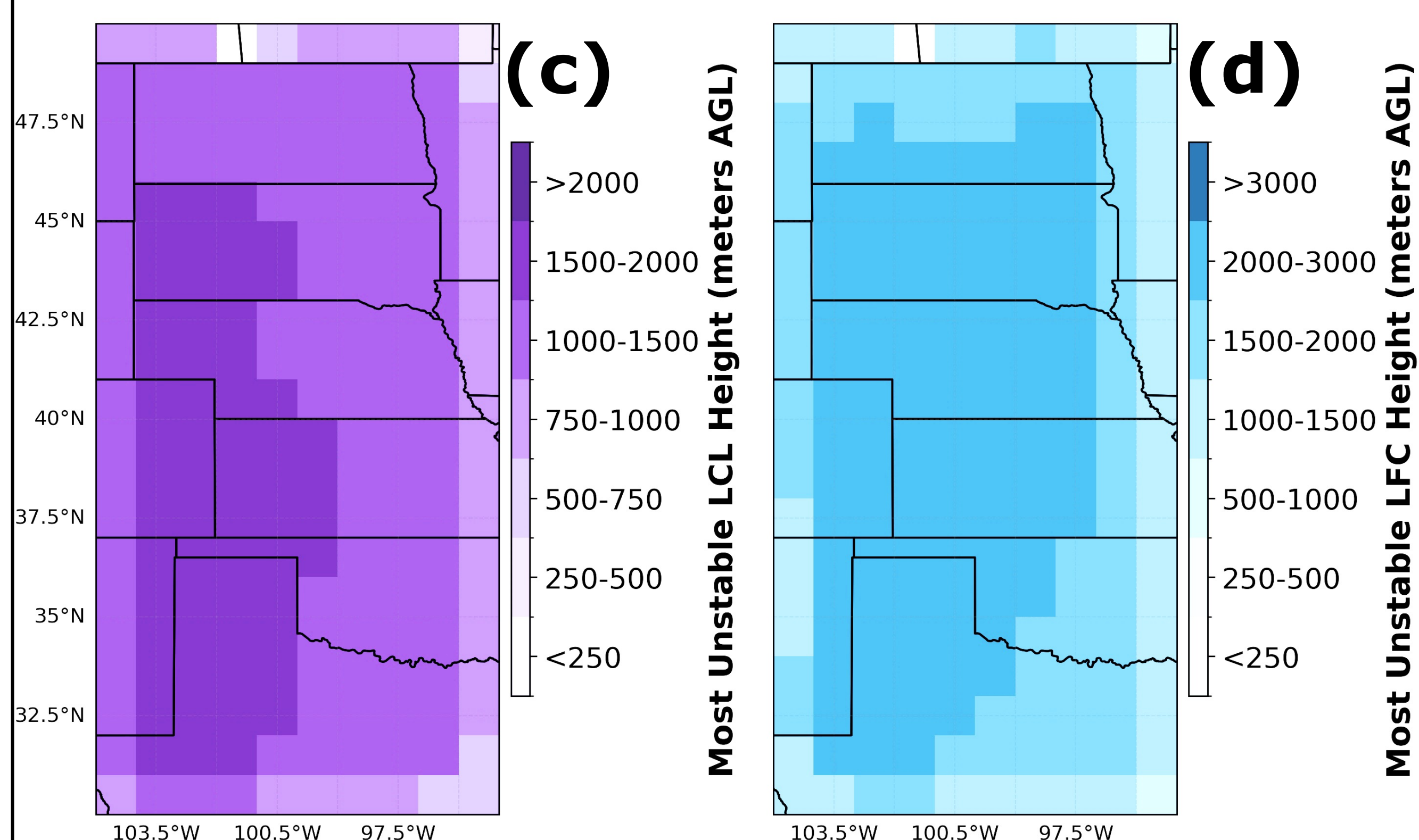


Fig. 2: 2003-2019 Average of Mean LCL Height (c) vs Most Unstable LFC Height (d)

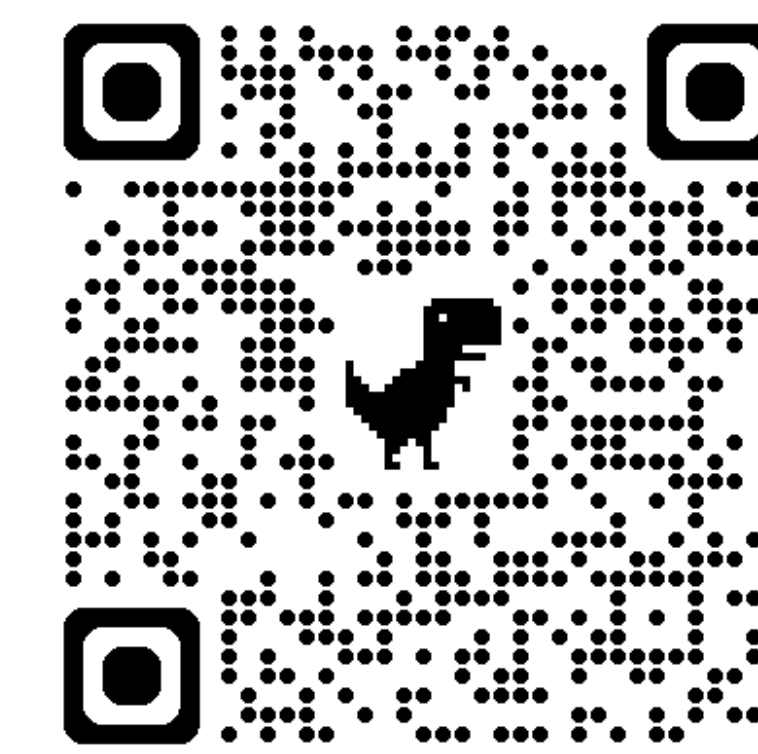
vi) Key Parameterizations and Results

1) Dependence of location, excessive CAPE, excessive shear, and lapse rates are **poorly represented in SHIP**

2) Some parameters in SHIP including 500 mb temp and MU mixing ratio **are not important** for hail size

3) ML identified **ECAPE, 6km Shear, and Freezing Level Height** as parameters to include in SHIP

4) Hail prediction will benefit from new parameterization



iii) Crucial Variables

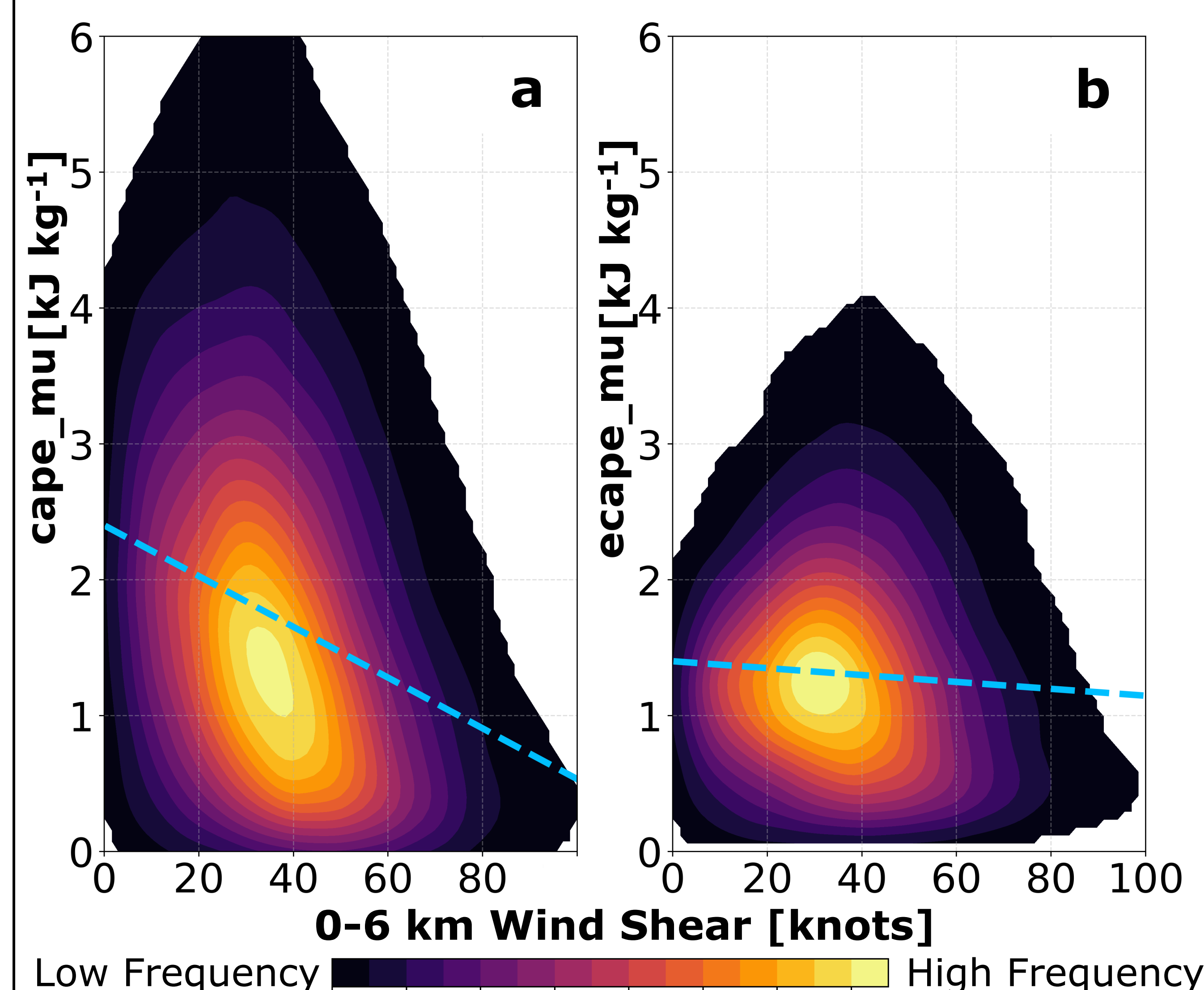


Fig. 3: Heatmap of MU_CAPE (a) and MU_ECAPE (b) vs 0-6 km wind shear. Blue lines show linear fits.

Acknowledgements

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iv) Simple SHIP and SHIP

Significant Hail Parameter (SHIP) variables vs **Simple SHIP variables**

- MU_CAPE [J kg^{-1}]
- MU_ECAPE [J kg^{-1}]**
- MU Mixing Ratio [g kg^{-1}]
- 700-500 mb Lapse Rate [K km^{-1}]**
- $(-1) \times 500 \text{ mb Temp. } [^{\circ}\text{C}]$
- 0-6 km shear [m s^{-1}]**

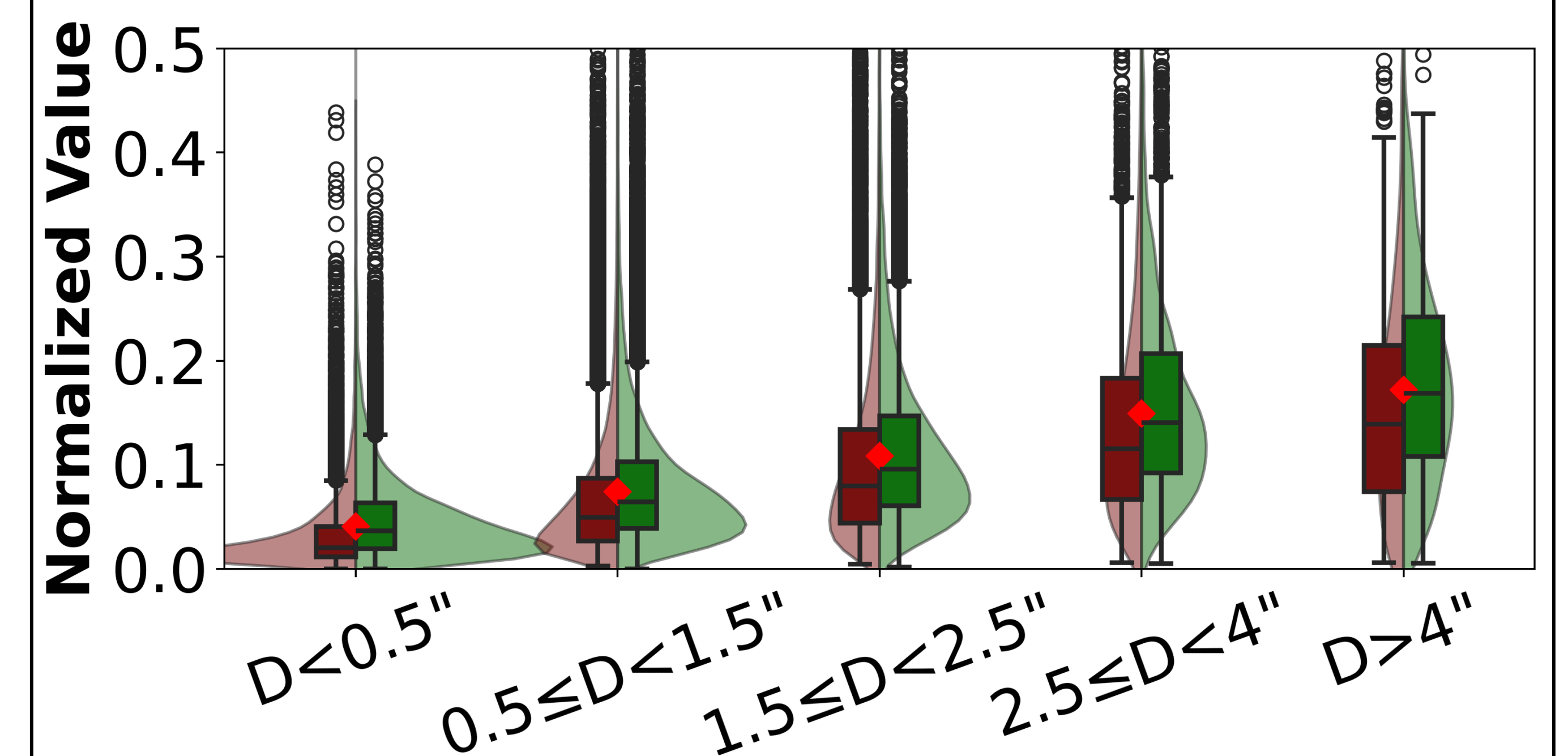


Fig. 4: SHIP (red) and Simple SHIP (green) vs hail size[in]

v) Machine Learning (ML) Validation

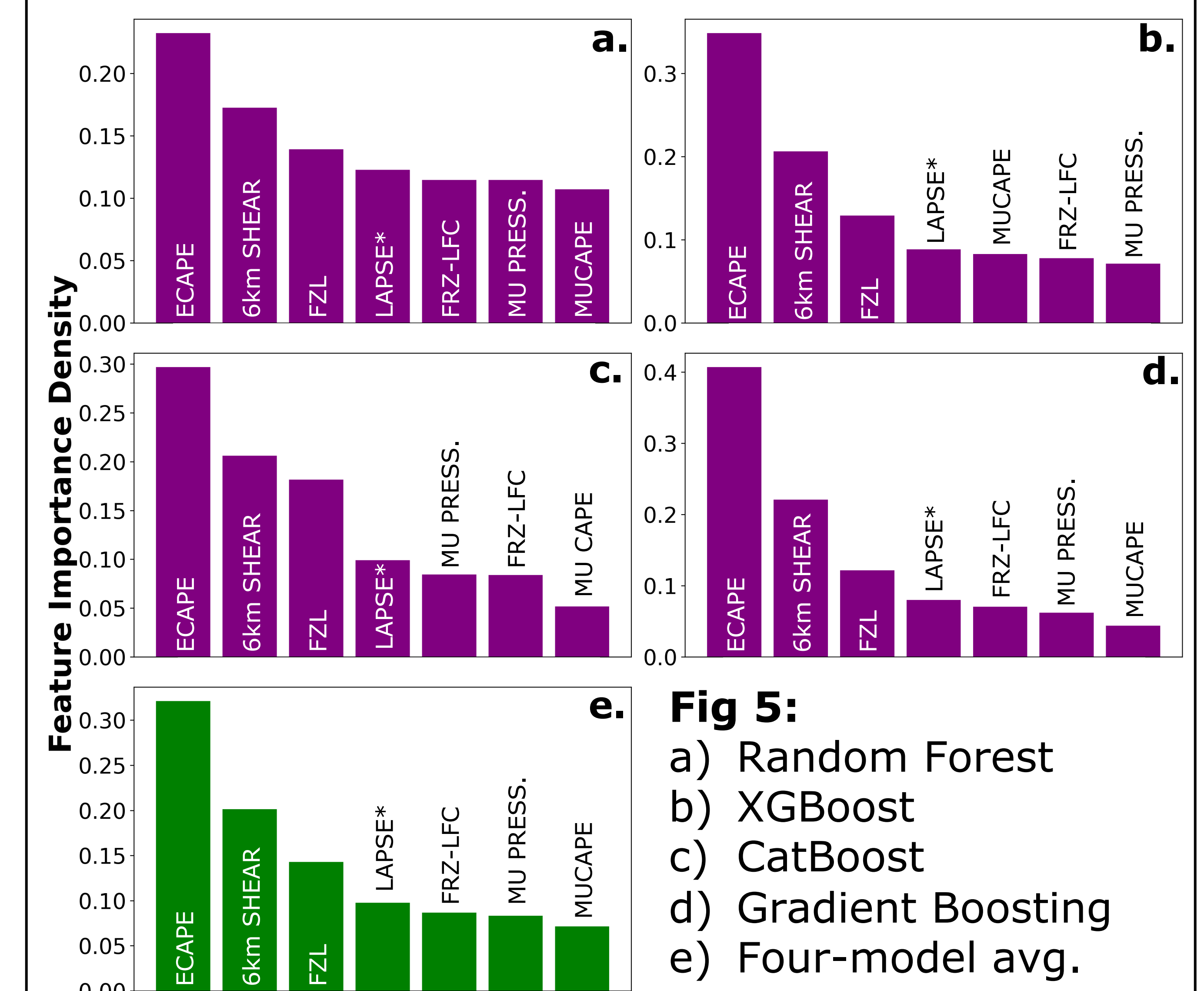


Fig 5:

- a) Random Forest
- b) XGBoost
- c) CatBoost
- d) Gradient Boosting
- e) Four-model avg.

* LAPSE refers to 500-700mb lapse rate

Using inputs determined by preliminary Simple SHIP results, four ML regression models agreed on **ECAPE, 6km Shear, and Freezing Level Height (FZL)** as the top three factors in hail size prediction.